LY VIREAK DARA

Computer Vision Researcher · Open-Vocabulary Detection · Detection Robustness · Multi-Object Tracking
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RESEARCH INTERESTS

I work on open-vocabulary object detection and what makes it break when the query wording changes. My interests are detection robustness under language variation, multi-object tracking, adapting models without retraining, and running them efficiently at the edge.

My current work looks at how open-vocabulary detectors fail when the prompt wording shifts. My benchmark *VocabDriftMOT* shows that recall drops sharply the moment a class word is swapped out, even for a close synonym, and that how near two words sit in CLIP space tells you nothing about whether detection survives the swap.

EDUCATION

M.S. in Artificial Intelligence, Xidian University, Xi'an, China *Sep 2023 – Jun 2026*
Thesis: *Enhanced Open-Vocabulary Object Detection using YOLO Architectures*. Defended May 2026; degree conferred 17 June 2026. GPA 3.4/4.0.

B.A. in Computer Science and Engineering, Royal University of Phnom Penh, Cambodia *Sep 2017 – Aug 2021*
Earned while working full-time.

PUBLICATIONS

Five peer-reviewed publications: one first-author (Scopus-indexed) and four co-authored across IEEE and SPIE venues. Author's name in bold.

First-author

[1] **V. D. Ly** (2025). EnYOLO-World: Enhanced Open-Vocabulary Detection with PGI and FiLM-PAN. *Electronic Letters on Computer Vision and Image Analysis (ELCVIA)*, Scopus-indexed. DOI: 10.5565/rev/elcvia.2441.

Co-authored

[2] R. Wang, C. Liu, L. Sun, L. Wang, **V. D. Ly**, C. Jiao (2025). Fine-Grained Meta-Learning with Semantic Augmentation for SAR Change Detection. *2025 International Joint Conference on Neural Networks (IJCNN)*, IEEE, pp. 1–8.

[3] R. Wang, C. Liu, L. Wang, L. Sun, **V. D. Ly**, C. Jiao (2025). Fine-Grained Meta-Learning with Semantic Augmentation for Synthetic Aperture Radar Change Detection. *Journal of Applied Remote Sensing*, 19(4), 046502, SPIE. [journal extension of [2]]

[4] X. Huang, R. Wang, L. Sun, I. Islam, **V. D. Ly**, C. Jiao (2025). Non-Rigid Registration Based on Rotation-Invariant Feature Flow. *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*.

[5] J. Chen, R. Wang, F. U. Maheri, **V. D. Ly**, C. Jiao, W. Li (2025). An Edge-Enhanced Siamese Network for 3D Change Detection. *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*.

RESEARCH EXPERIENCE

VocabDriftMOT: Vocabulary-Drift Robustness Benchmark for Open-Vocabulary Detection *2025 – 2026*
Lead researcher. Manuscript in preparation (arXiv). github.com/Vireakdara/VocabDriftMOT

- Built the first benchmark for vocabulary-drift failures in open-vocabulary detection, on YOLO-World: 49 term-shift events across 7 MOT17 sequences, measured over 30-frame windows.
- Found that recall collapses when a class word is swapped. "Pedestrian" on its own scores about 0.02, but adding "person" back brings it to 0.84. CLIP similarity does not explain this: "human" is the closest word to "person" (cosine 0.91) yet still scores 0.00.
- Wrote the transition-generation and verification pipeline, and a recall metric (Recall_AT) that catches these failures where the usual detection metrics show nothing wrong.

EnYOLO-World: Open-Vocabulary Detection Architecture *2024 – 2025*
First author; published in ELCVIA.

- Added a PGI backbone and a FiLM-conditioned PAN to YOLO-World, gaining +3.1 AP over baseline (52.2 AP on COCO at 78 FPS), and ran ablations to check what each part contributed.

Applied ML systems (selected) *2025 – 2026*

- Built a multimodal RAG pipeline (CLIP embeddings, a ChromaDB store, a cross-encoder reranker and LLaMA3, with Qwen2.5-VL-72B writing the image captions), and deployed it on GCP Cloud Run with FastAPI and Docker.
- Built OpenDet-Bench, an open-vocabulary detection and tracking service (YOLOv8 + ByteTrack + ONNX, 1.54× faster inference, MLflow), running real-time video analytics on GCP Cloud Run.

TECHNICAL SKILLS

Deep learning: PyTorch, TensorFlow, YOLO (v5–v11), CLIP, Hugging Face Transformers, Scikit-learn

Vision & VLMs: object detection, multi-object tracking, ByteTrack, OpenCV, TorchVision, ONNX; Qwen2.5-VL, LLaMA3, LangChain, FAISS, ChromaDB

Evaluation & MLOps: benchmark construction, custom metrics, annotation pipelines, MLflow, Docker, GCP Cloud Run, CI/CD (GitHub Actions), Git

Programming & languages: Python, Java, C/C++, JavaScript/TypeScript, SQL · English (fluent), Chinese (HSK 3), Khmer (native)

PROFESSIONAL EXPERIENCE

Software Engineer, KOSIGN (Korea Software Innovation Global Network) *Oct 2020 – Aug 2023*

- Built and ran full-stack enterprise applications (Java, Spring Boot, MVC) for users across Korea and Southeast Asia, designed and tuned the PostgreSQL schemas and REST APIs behind them, and worked day to day on a fully English-speaking Agile team.

Software Engineer, Web Essentials *Oct 2019 – Aug 2020*

- Built responsive React.js/TypeScript frontends and wired in third-party payment and authentication APIs, with tests to keep the integrations honest.